

Predicting House Prices Using Real-Estate Features

DSK807: Applied Machine Learning

The report is prepared by:

Liva Igaune
2003200, Data Science, 3d semester

Erika Laizane
09121990, Data Science, 3d semester

Rabinson Pariyar
09111998, Data Science, 3d semester

Lecturer:
Tariq Yousef

Department of Mathematics
and Computer Science

Table of Contents

1. Introduction	3
2. Dataset & EDA.....	3
3. Data Preparation	3
3.1 Feature Screening and Selection.....	3
4. Model Performance.....	4
4.1.Prediction Error Analysis	4
4.2. Model Comparison	5
4.3. Interpretability and SHAP Analysis	5
5. Neighbourhood Effects.....	5
9. Conclusion	5

1. Introduction

This project explores multiple machine learning approaches on the Ames Housing dataset, ranging from linear models to ensemble methods and neural networks. The objective is to evaluate predictive performance, analyse feature importance, and understand which modelling techniques are most suitable for structured tabular data.

2. Dataset & EDA

The Ames Housing dataset contains residential property sales in Ames, Iowa, with 79 explanatory variables that describe structural, locational, and quality-related attributes. The target variable is the Sale Price of houses. Features include numerical variables (e.g., living area, basement size, year built) and categorical variables (e.g., neighbourhood, quality ratings).

Preprocessing: We applied a logarithmic transformation to the Sale Price to correct its right-skewed distribution and handle outliers.

Feature Relationships: Exploring the Sale Price relationships with square footage, number of rooms, and location features uncovered interesting insights. The living area shows a strong relationship with the price, with increasing variance for larger homes and a small number of extreme outliers. Neighbourhood exhibits substantial price differentiation, confirming strong location effects not captured by simple numeric correlations. Sales price increases with the number of rooms up to a threshold, after which it marginally drops.

Correlation: Correlation heatmap revealed that size and quality-related variables dominate linear relationships with the target Sales Price, while many features exhibit weak or nonlinear correlations.

3. Data Preparation

Missing values were handled using feature-specific strategies, for example: filling the median value for 'Lot Frontage', filling zeros for 'Bsmt Full Bath' and 'Mas Vnr Area' and assigning zero where 'Garage Yr Blt' was missing. Additionally, the remaining categorical variables were one-hot encoded, resulting in 260 features.

3.1 Feature Screening and Selection

Feature screening was performed using a combination of correlation with the target and Random Forest feature importance. Features were kept if they exceeded either a correlation threshold of

0.3 or a Random Forest importance threshold of 0.003. This resulted in a final feature set of 51 variables.

4. Model Performance

Linear Regression: A strong baseline with surprisingly high metrics, indicating successful feature selection that was able to linearise the relationship.

Random Forest: In this application, even after hyperparameter tuning, Random Forest performance was slightly inferior to the other shallow models evaluated.

Gradient Boosting: Gradient Boosting achieved better performance than Random Forest and Linear Regression. The effect of hyperparameter tuning was minimal however it was the push that enabled the performance to surpass other models.

Ensemble Learning: A Stacking Ensemble combining Random Forest, and Gradient Boosting as base learners with Linear Regression as a meta-learner was constructed. The performance was identical to the Gradient Boosting, indicating that the ensemble does not add additional predictive power, and Gradient Boosting is already capturing most of the useful structure in the data.

Feed-Forward Neural Networks: In the process of improving the feed-forward base model, multiple neural network configurations were evaluated, including different optimizers (Adam, RMSprop, SGD), learning rates, activation functions, regularization techniques (dropout, L2) and network width variations. The best neural network achieved competitive performance but did not outperform tree-based ensemble models.

Autoencoder: Autoencoders were trained to learn compressed latent representations of the feature space. While reconstruction loss decreased with training, the Random Forest model using latent features underperformed compared to models trained on the original feature space. This suggests that dimensionality reduction via autoencoders did not preserve the most predictive information for this task.

4.1. Prediction Error Analysis

To further investigate the model performance, we analysed mean absolute error (MAE) across four different price ranges (low, mid-low, mid-high, high) for all models. This analysis enables a direct comparison of how each model performs across different segments of the housing market and helps identify where certain models struggle or excel. From the results, we observe that all models have higher errors for low-priced houses, and the best performance is in mid-range

prices. The Stacking Ensemble and Gradient Boosting consistently achieve the lowest mean absolute error across all price ranges, indicating strong generalisation.

4.2. Model Comparison

Model	RMSE	MAE	R ²
Linear Regression	0.111	0.079	0.925
Random Forest	0.124	0.084	0.907
Gradient Boosting	0.111	0.077	0.926
Stacking Ensemble	0.111	0.077	0.926
FFNN	0.123	0.090	0.908
RF + Autoencoder	0.147	0.104	0.869

4.3. Interpretability and SHAP Analysis

SHAP analysis of Random Forest and Gradient Boosting models confirms that overall quality and living area are the dominant drivers of house prices, followed by basement size, garage capacity, and construction year. Both models agree on the direction and magnitude of these effects. Gradient Boosting concentrates more heavily on the strongest predictors, while Random Forest distributes importance across a wider set of features.

5. Neighbourhood Effects

Although exploratory analysis suggests that neighbourhood influences house prices, explicit neighbourhood variables were not among the most important predictors after feature selection. Correlation-based filtering and Random Forest importance favoured structural attributes such as quality, size, and age, which are themselves strongly correlated with location. SHAP analysis confirms that models primarily rely on these attributes rather than explicit neighbourhood labels. This indicates that neighbourhood effects are captured indirectly through correlated housing characteristics like year built, overall quality, basement size, and living area.

9. Conclusion

This project evaluated multiple machine learning models for house price prediction using the Ames Housing dataset, including linear regression, ensemble tree models, neural networks, autoencoders, and a stacking ensemble. The results show that model performance depends more on preprocessing and feature engineering than on model complexity.

Gradient Boosting, Linear Regression, and the Stacking Ensemble achieved very similar performance, with RMSE values around 0.11, MAE near 0.08, and R^2 close to 0.92. Given the identical performance, Gradient Boosting was selected as the final model due to its lower complexity and easier interpretability compared to the ensemble. This indicates that once informative features are selected, simpler models can be nearly as effective as more complex ensembles.

Even after huge improvement from the base model FFNN and autoencoder lagged behind tree-based models. This aligns with known limitations of deep learning on structured tabular data.

Feature importance analysis using correlation, Random Forest importance, and SHAP values consistently identified overall quality, living area, basement size, garage capacity, and construction or renovation year as the strongest drivers of house prices. Although exploratory analysis showed clear price differences across neighbourhoods, neighbourhood variables had a limited direct impact in the final models, indicating that their effects are largely captured indirectly through correlated structural and quality features.

Future work for improved accuracy could focus on geographic refinement using GPS coordinates instead of neighbourhood labels, incorporate external data such as interest rates and economic indicators and using specialized models for low priced segment to address the higher error rates.